CropShield AI Evaluation Report

Saved evaluation evidence and explicitly pending results

Evidence and scope

This report summarizes existing local JSON artifacts inspected on 10 October 2026. It does not represent a new run unless explicitly stated. Saved RAGAS/DeepEval live judge score artifacts were not found; these provider-dependent metrics are pending successful execution with a reachable, funded compatible endpoint.

Retrieval benchmark (saved run)

Metric	Saved value	Interpretation
Benchmark	20 labeled queries; timestamp 2026-10-09 18:49:30 UTC	Historical saved benchmark; not rerun for this document.
Precision@1 / @3 / @5	0.9 / 0.85 / 0.83	Crop-label precision at each cutoff.
Recall@5	0.4183	Fraction of expected relevant crop labels retrieved by rank 5.
MRR	0.925	Mean reciprocal rank of first relevant result.
Hit rate@5	0.95	Share of queries with at least one relevant result among top five.
Duration	1.02 seconds	Recorded duration for the saved run.

Machine learning (saved held-out split)

Model / measure	Value	Evaluation set
RandomForestRegressor R^2	0.9876	300 held-out samples
Yield MAE	0.4992	Yield target units follow the dataset; synthetic data.
Yield RMSE	1.6865	Same held-out split.
RandomForestClassifier accuracy	0.5533	300 held-out samples
Suitability precision / recall / F1	0.5086 / 0.5533 / 0.4737	Macro/aggregate behavior depends on saved evaluation implementation.

LLM evaluation

RAGAS: pending. DeepEval: pending. The saved evaluation UI/runtime reported judge provider connectivity/credit errors; per-query RAGAS and DeepEval scores must not be inferred from retrieval ranking metrics. Current saved report artifacts include an older aggregate judge summary dated 2026-10-08 14:14:15 UTC with overall score 0.972 and groundedness 1.0. This is retained as a historical artifact only; its provider configuration and run details were not independently revalidated here.

Tests

Verification run on 10 October 2026: 30 tests passed. Pytest emitted one non-blocking cache write warning because the sandbox denied access to its cache nodeids file. Re-run the command in README.md for a fresh result.

How to interpret

- Retrieval labels measure crop-label matching, not expert-validated relevance or answer correctness.
- The dataset is synthetic; its metrics are development signals, not evidence of field accuracy.

- Run a fresh local retrieval benchmark and a funded/reachable LLM judge before presenting live RAGAS or DeepEval scores.